

SCALE 1:100

FREQ. REF. 1GHz

THE PHYSICAL LAYER

An engineering deep dive into how copper, fiber, and radio waves bind the digital world to physical reality.

IMPEDANCE: 106Ω
LENGTH: 50m (MA3)

10110

DATA RATE: 10 Sbps
BER TARGET: $< 10^{-12}$

PHYSICAL MEDIA DEPENDENT (PMD)
INTERFACE SPECIFICATIONS

DIPOLE ARRAY

COVERAGE RADIUS: 100m

FSPL: 92 dB @ 100m
POWER: 12dBm
BW: 20 MHz
MODULATION: QAM

DIPOLE ARRAY
RF FRONT-END

ATTENUATION: 0.5 dB/km

POLARIZATION: VERTICAL

RF FRONT-END

LASER DIODE EMISSION
PHOTO DETECTOR INPUT

λ : 1310 nm (SINGLE MODE)



n_{core} : 1.447
 n_{cladding} : 1.444

COPPER CABLE

V_{pp}: 3.3V
RISE TIME: 100ps
DUTY CYCLE: 50%

+1.2V
3.3V
-8.2V
0V

IMPEDANCE: 106Ω

GND: 0V

LENGTH: 50m (MA3)

+1.2V THRESHOLD

0.6V

2.0V

1.0ps

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The Logical Bridge Between Bits and Physics

A bit is a logical abstraction; it cannot travel through a cable. Higher layers (IP, TCP) are entirely blind to the transmission mechanism. If the physics fail, the network fails.

IP ADDRESSES

192.168.1.1
192.168.1.1
10.0.0.1

TCP PORTS

80 (HTTP) / 443 (HTTPS) / 22 (SSH)

MAC ADDRESSES

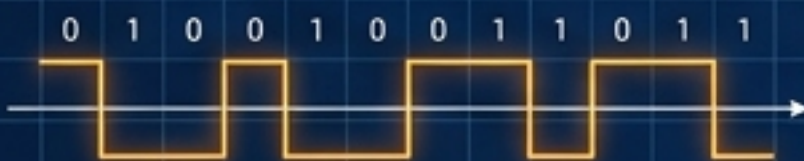
00:1A:2B:3C:4D:5E
FF:EE:DD:CC:4B:AA
FF:EE:DD:CC:BB:AA

THE LOGICAL WORLD

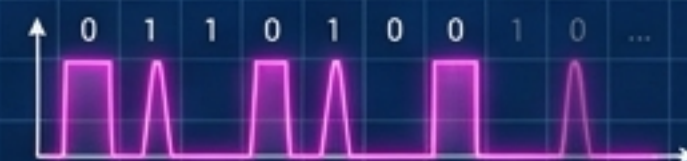
THE PHYSICAL WORLD



COPPER: ENCODED AS VOLTAGE CHANGES.



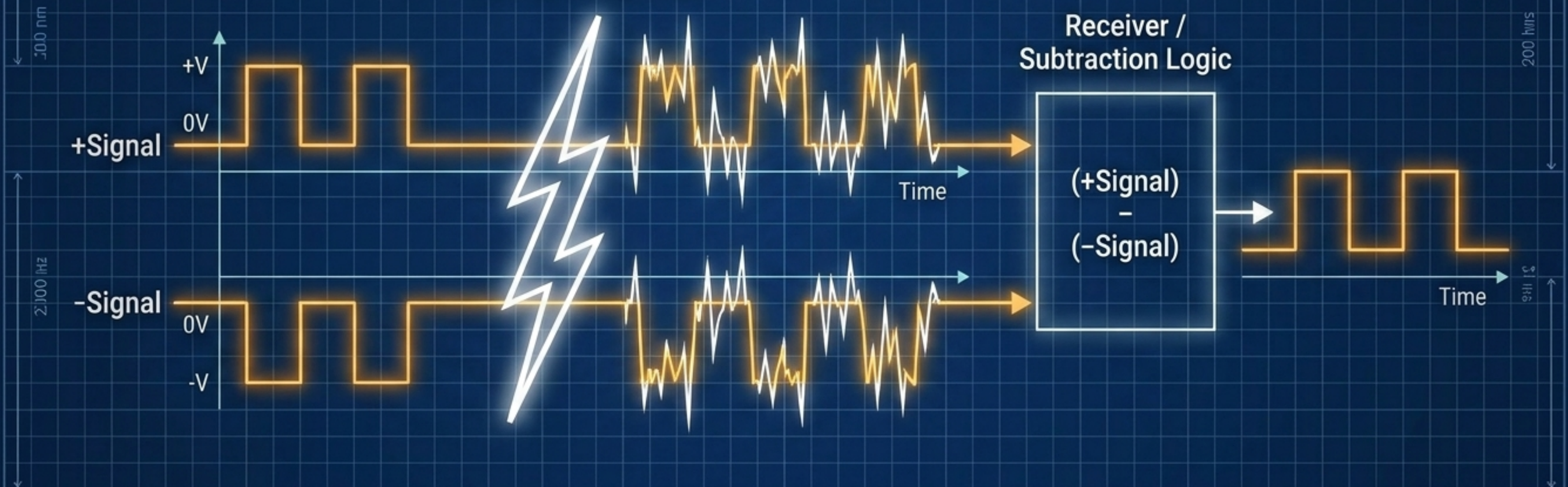
FIBER: ENCODED AS LIGHT INTENSITY.



WIRELESS: ENCODED AS PHASE/FREQUENCY SHIFTS.

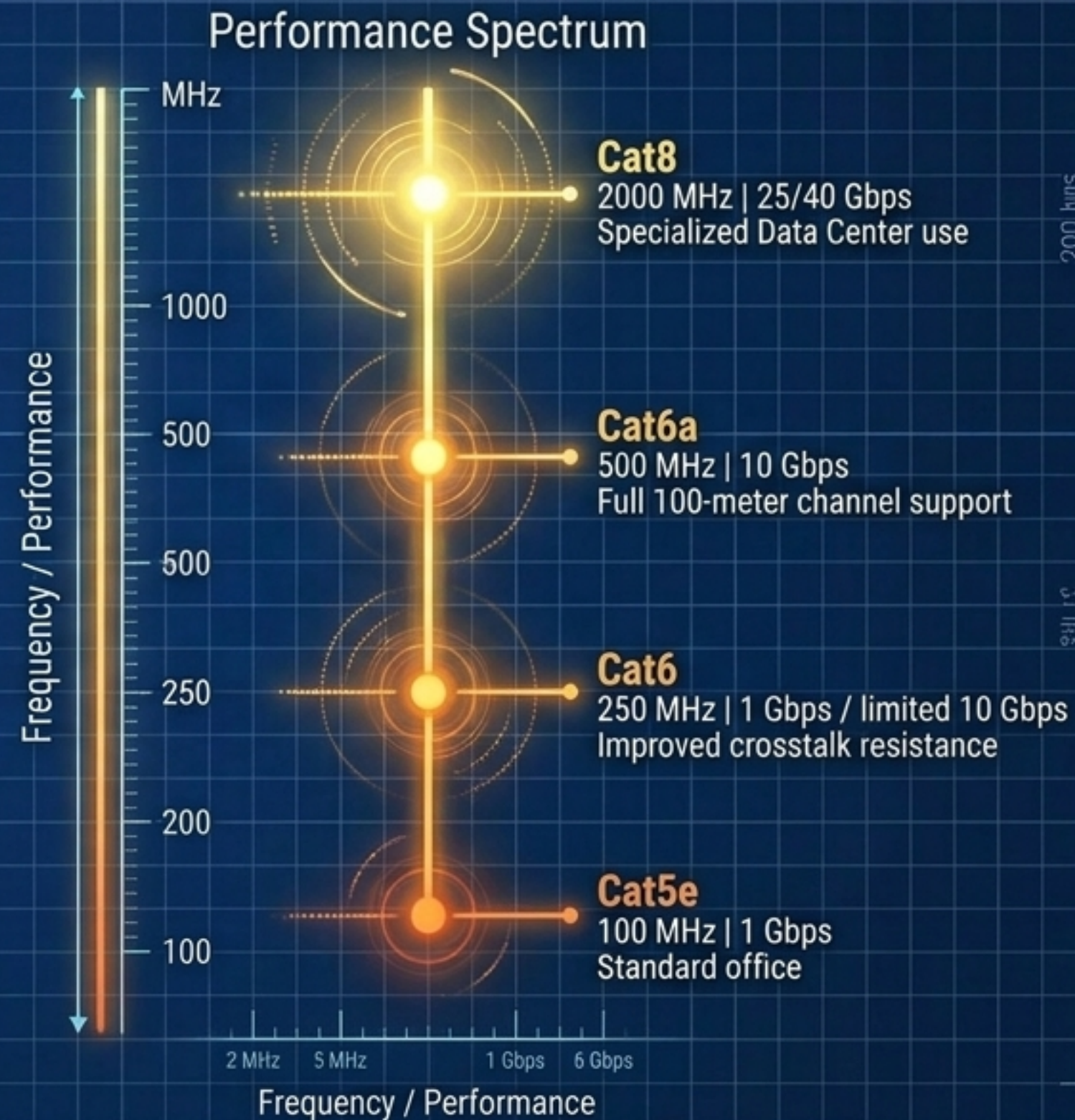
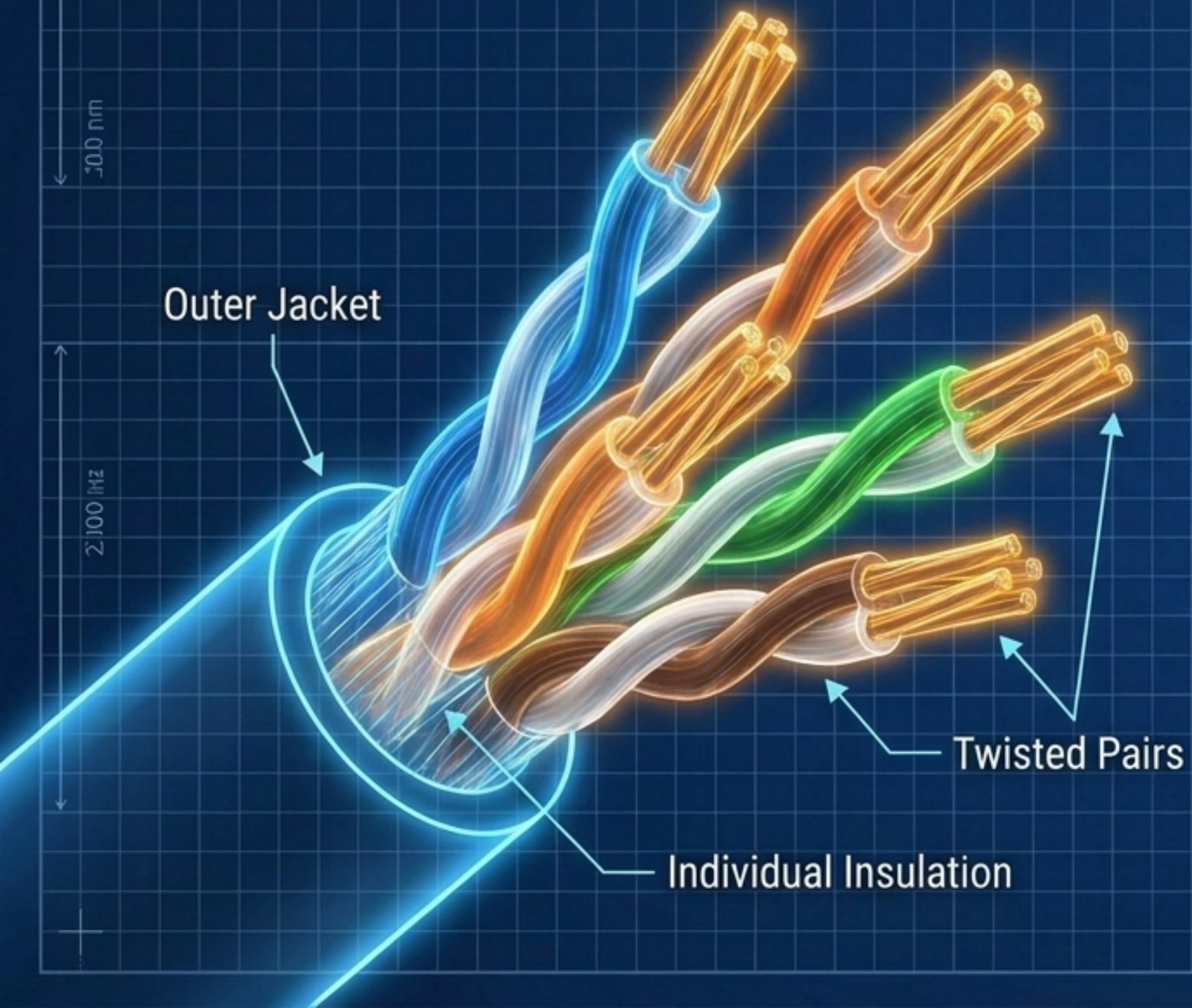


measure absolute voltage, it measures the difference between the two wires, elegantly rejecting common interference.



The Ethernet Cable is an Engineered Transmission System

Electrical signals degrade over distance (attenuation). The standard Ethernet channel limit is 100 meters (90m horizontal + 10m patch cords) to ensure the receiver can reliably distinguish the signal from the noise floor.

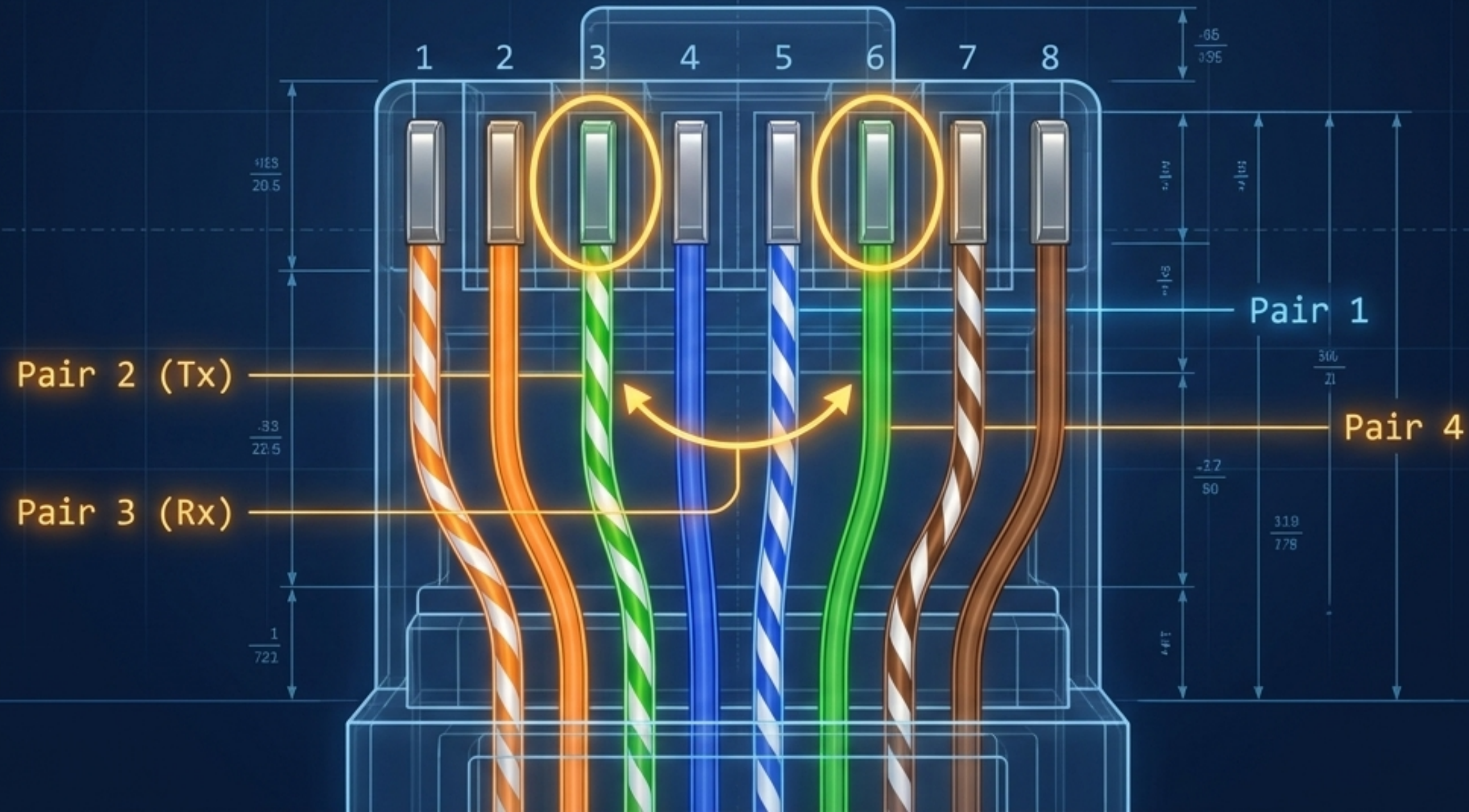


Terminating Copper: The T-568B Standard

Straight-Through: T-568B on both ends.

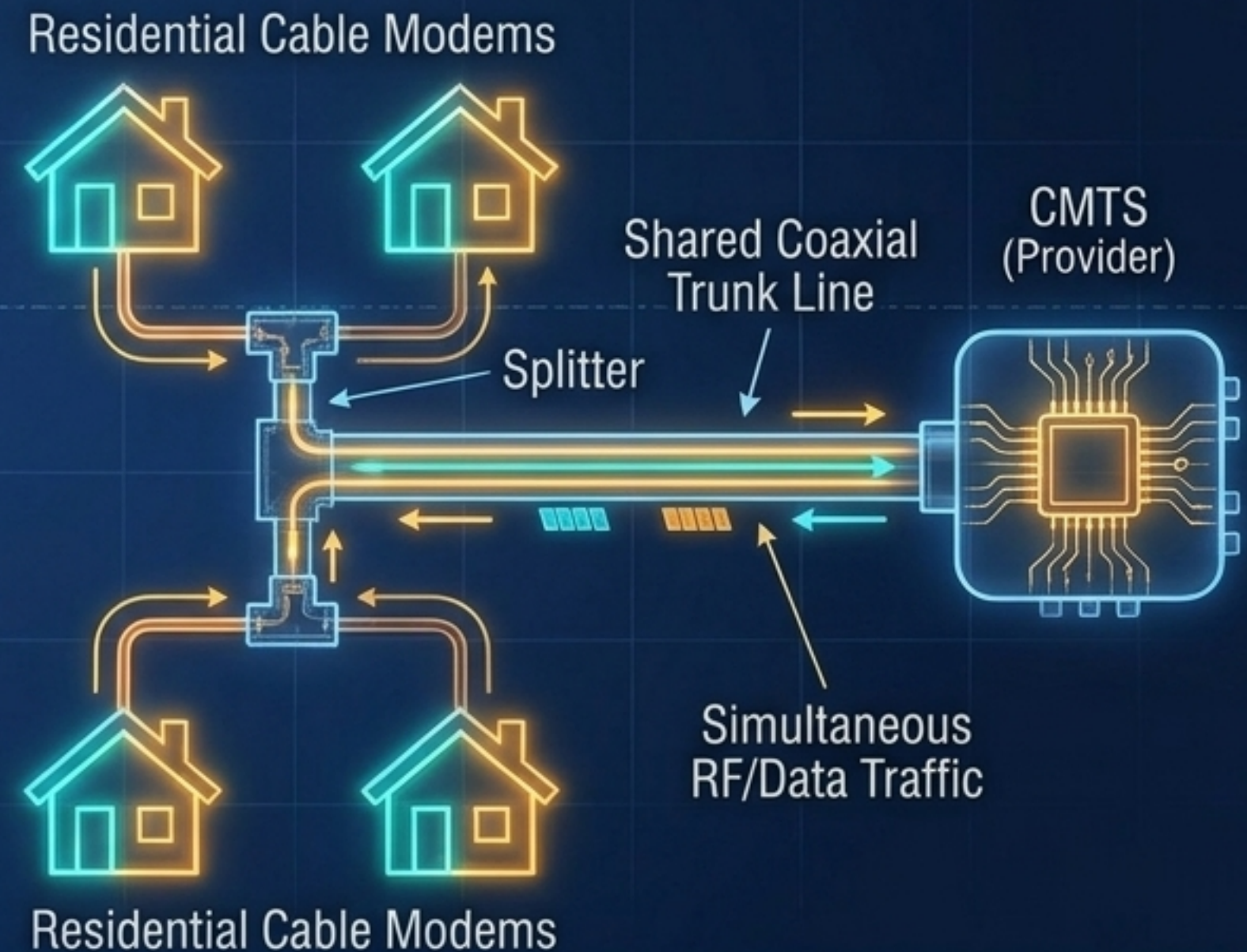
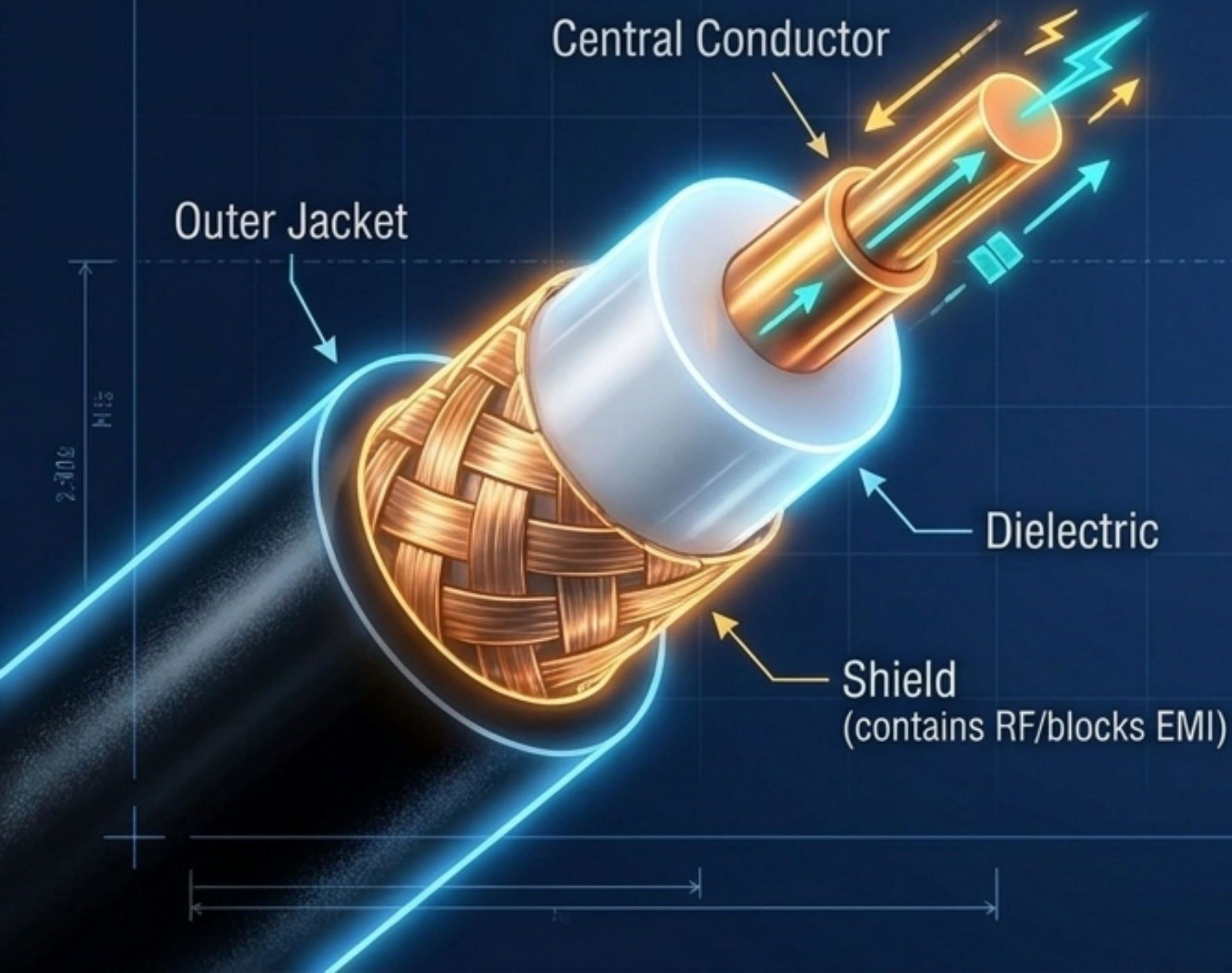
Crossover: T-568A on one end, T-568B on the other (crossing transmit and receive).

The Modern Fix: Auto-MDI/MDI-X allows modern equipment to automatically compensate for pair configuration, but incorrect physical pin orders still destroy signal integrity.



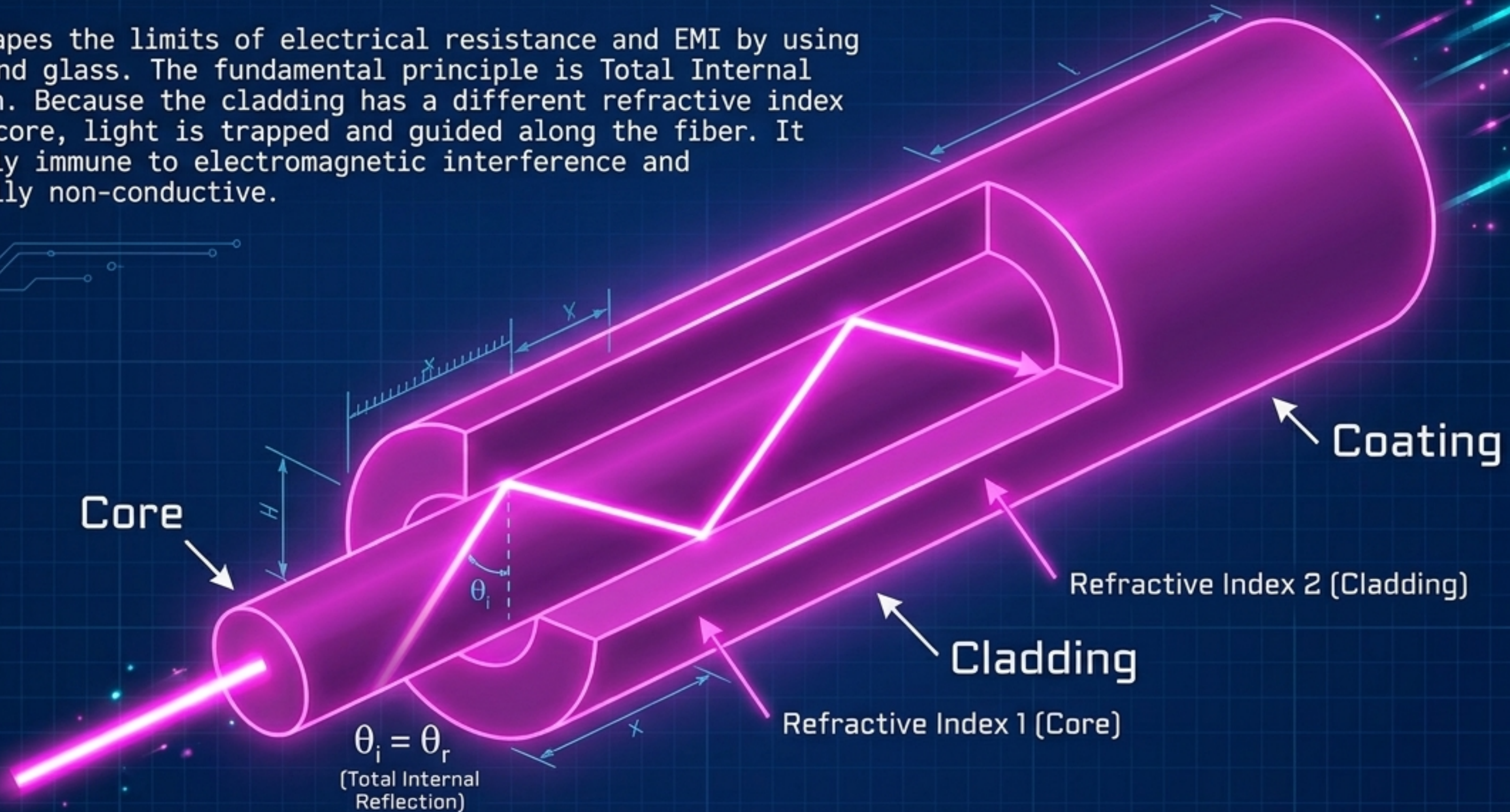
Coaxial Cable and Shared Media Architecture

Coax is heavily associated with RF distribution and DOCSIS broadband. Unlike an office Ethernet link, coax access networks shared medium. The physical infrastructure is shared among multiple subscribers, requiring the provider to coordinate traffic logically and manage the finite RF spectrum.



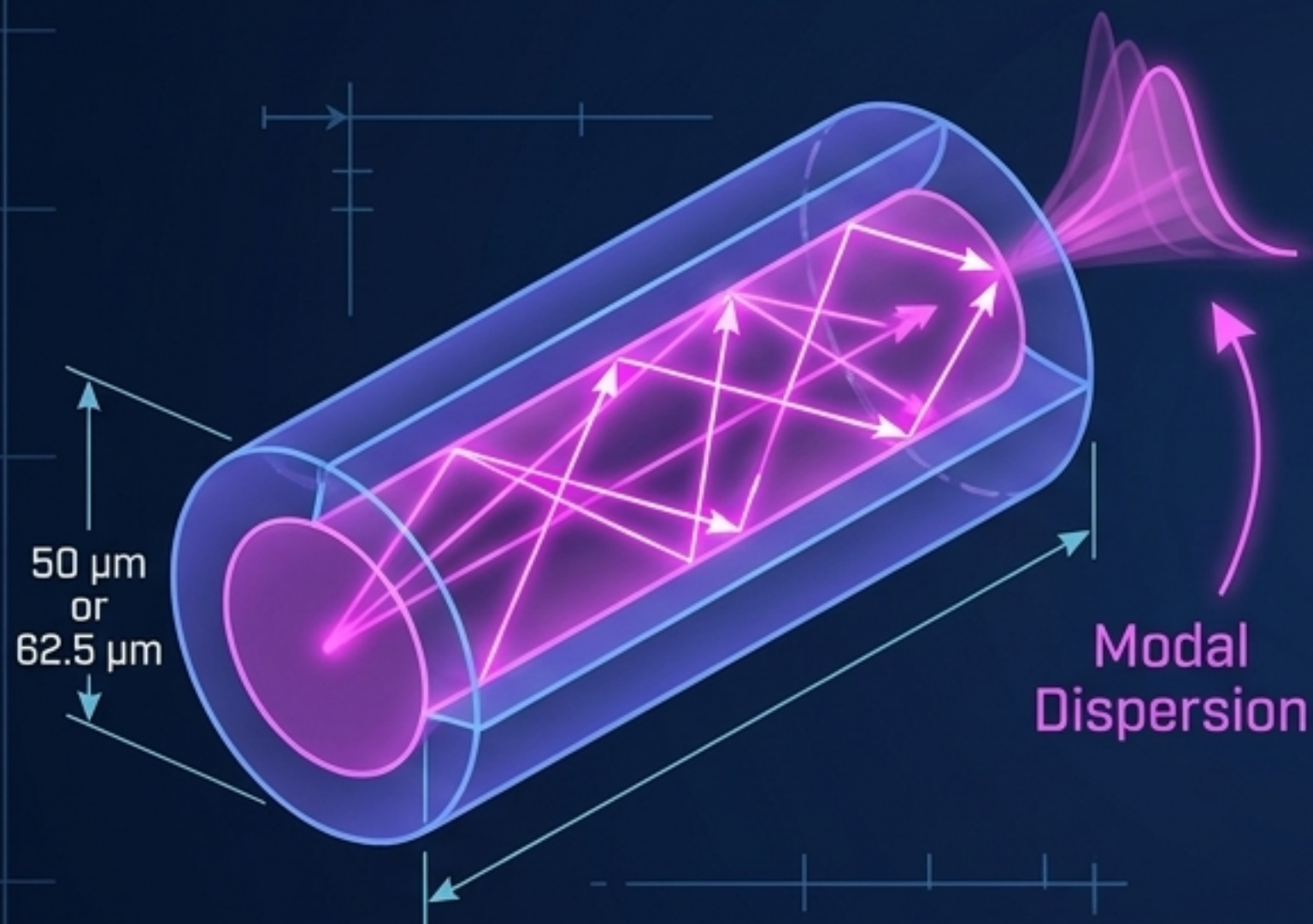
Fiber Optics: Engineering with Light

Fiber escapes the limits of electrical resistance and EMI by using photons and glass. The fundamental principle is Total Internal Reflection. Because the cladding has a different refractive index than the core, light is trapped and guided along the fiber. It is entirely immune to electromagnetic interference and electrically non-conductive.

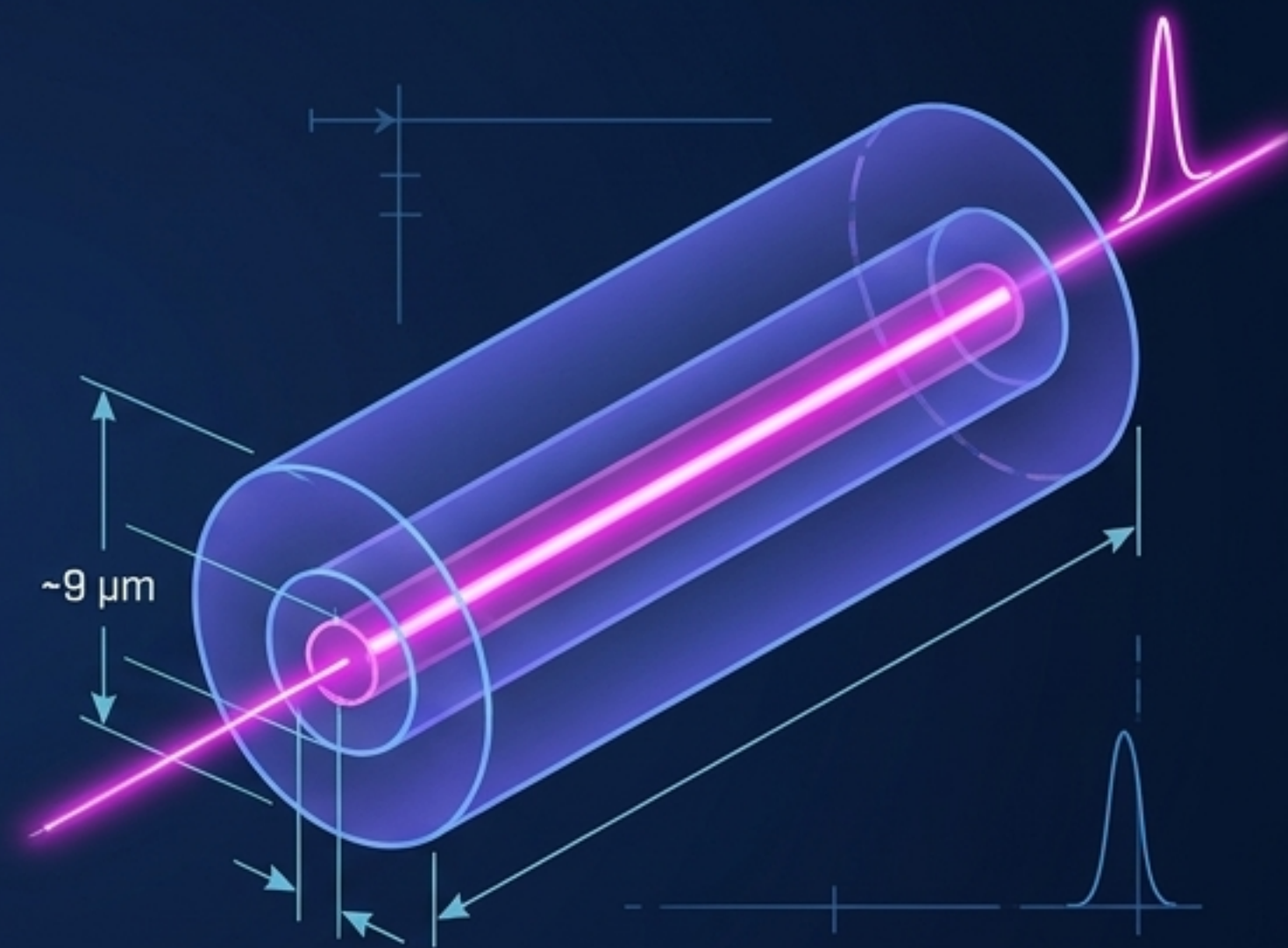


Managing Modal Dispersion: Single-Mode vs. Multimode

Multimode (MMF)



Single-Mode (SMF)



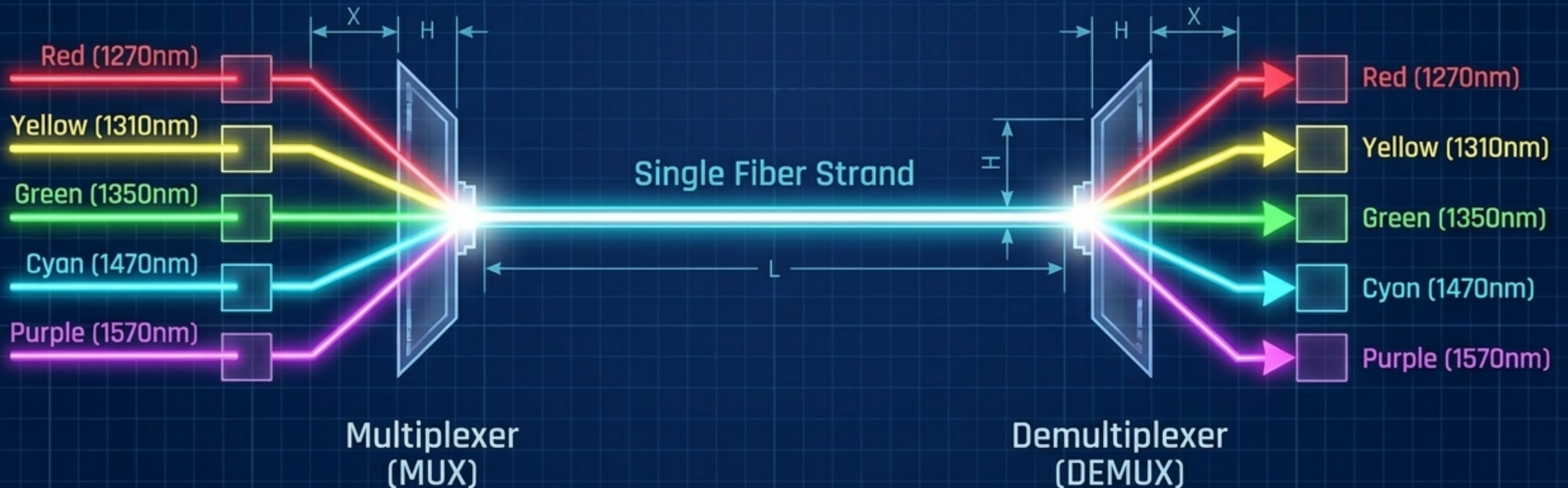
Reach:	~550m (OM3/OM4 at 850 nm)
Typical Use:	LAN, Data Centers

Reach:	100+ km (1310/1550 nm)
Typical Use:	WAN, ISP Backbone, Submarine

Multiplexing the Light

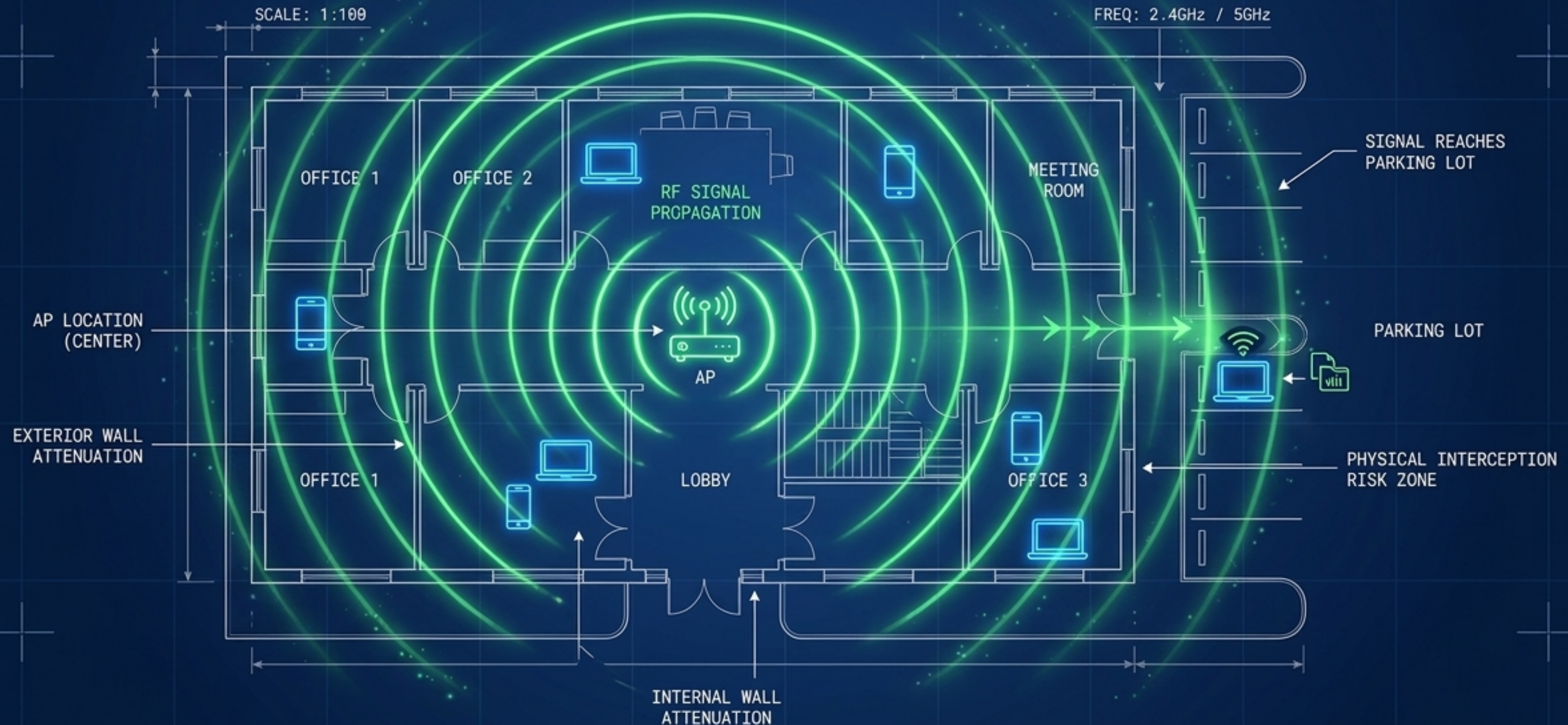
A fiber strand does not automatically have unlimited bandwidth. To multiply capacity without trenching new cable, engineers use Wavelength Division Multiplexing (WDM).

- CWDM (Coarse): Wavelengths are widely spaced.
- DWDM (Dense): Wavelengths are tightly packed, supporting massive channel counts for telecommunications and backbone networks.



Wireless: The Unbound Medium

Radio turns the physical environment itself into a shared medium. Because transmissions propagate through free space, physical isolation is impossible. An attacker only needs to be in range to capture frames. Therefore, strong encryption (WPA2/WPA3) is not an optional feature—it is a mandatory Layer 1 security requirement.

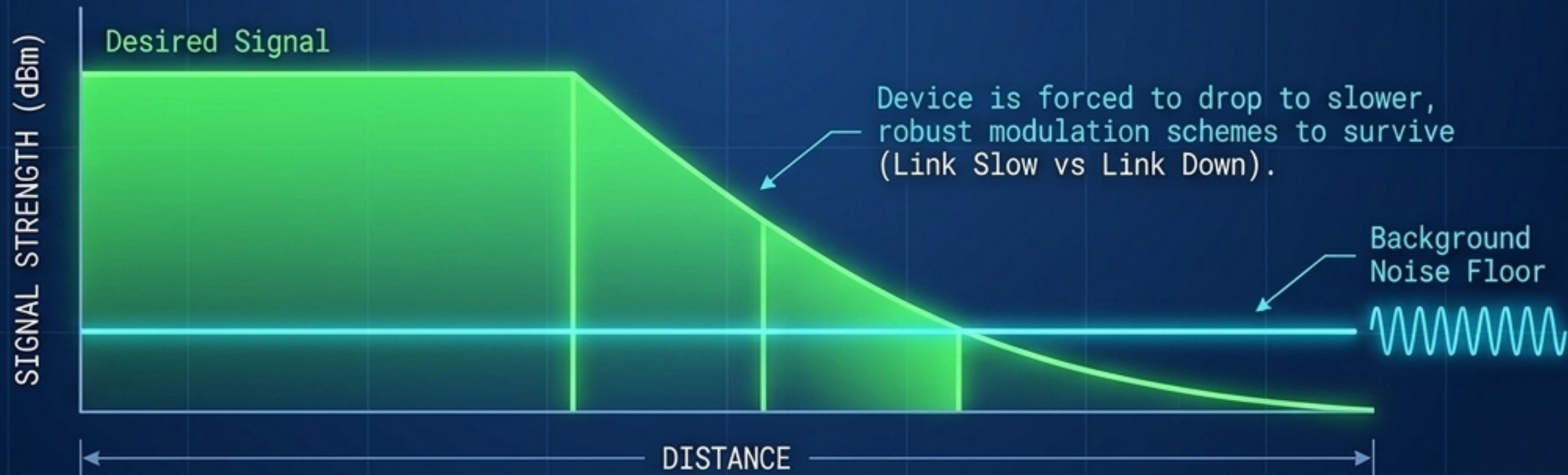


The Wireless Spectrum: 2.4 GHz vs. 5 GHz

Note: Frequency alone does not equal speed.

2.4 GHz	5 GHz
Superior wall penetration, longer range, highly congested spectrum (Bluetooth, microwaves).	High attenuation through obstacles, shorter range, massive usable spectrum, high potential throughput.

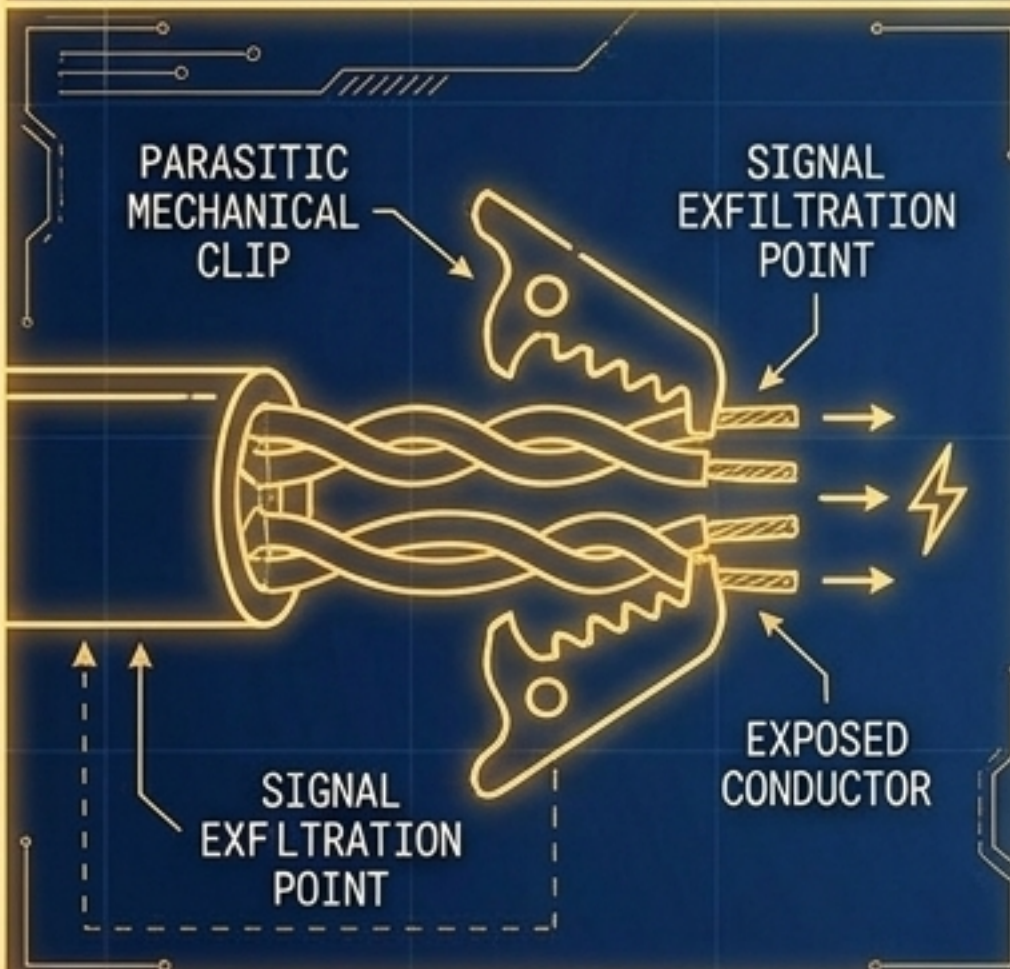
Signal-to-Noise Ratio (SNR)



The Physical Security Boundary

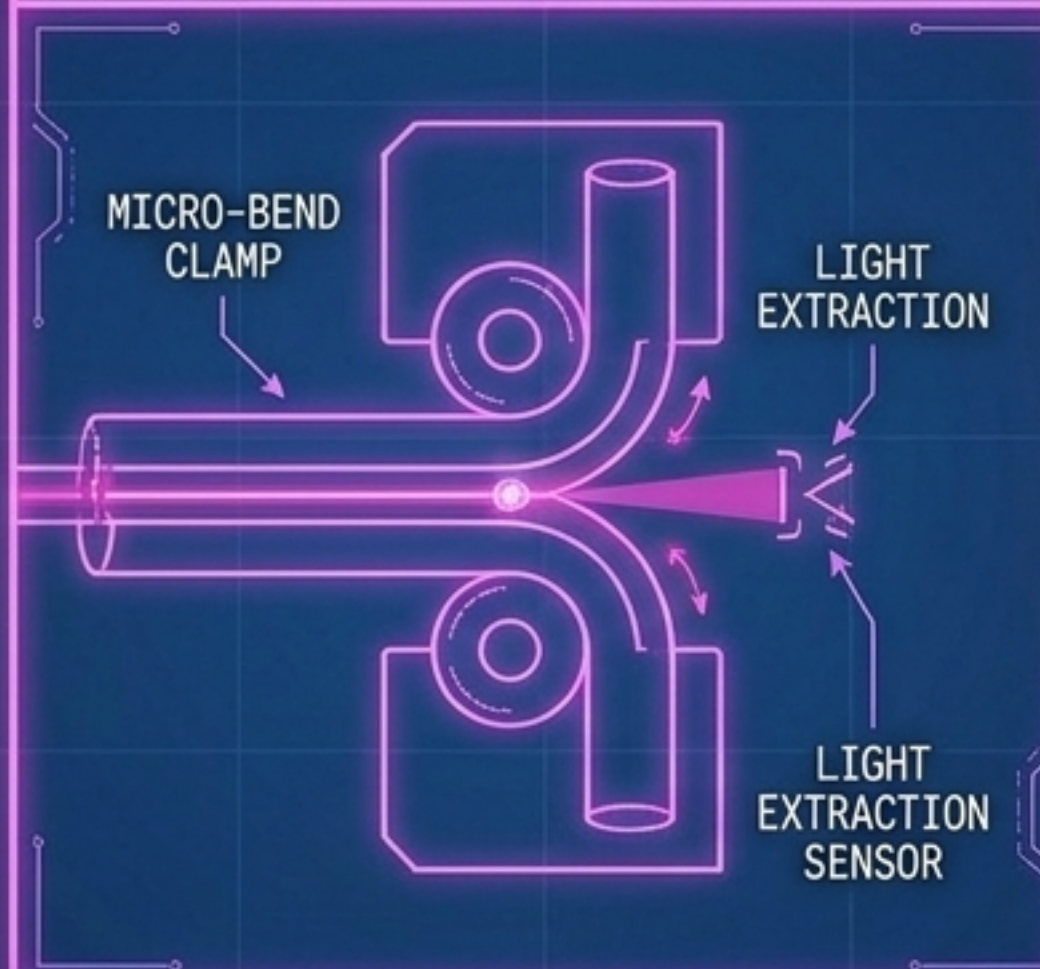
Logical security is bypassed if the physical layer is compromised. Network security begins below the application layer.

Copper & Coax (Electrical)



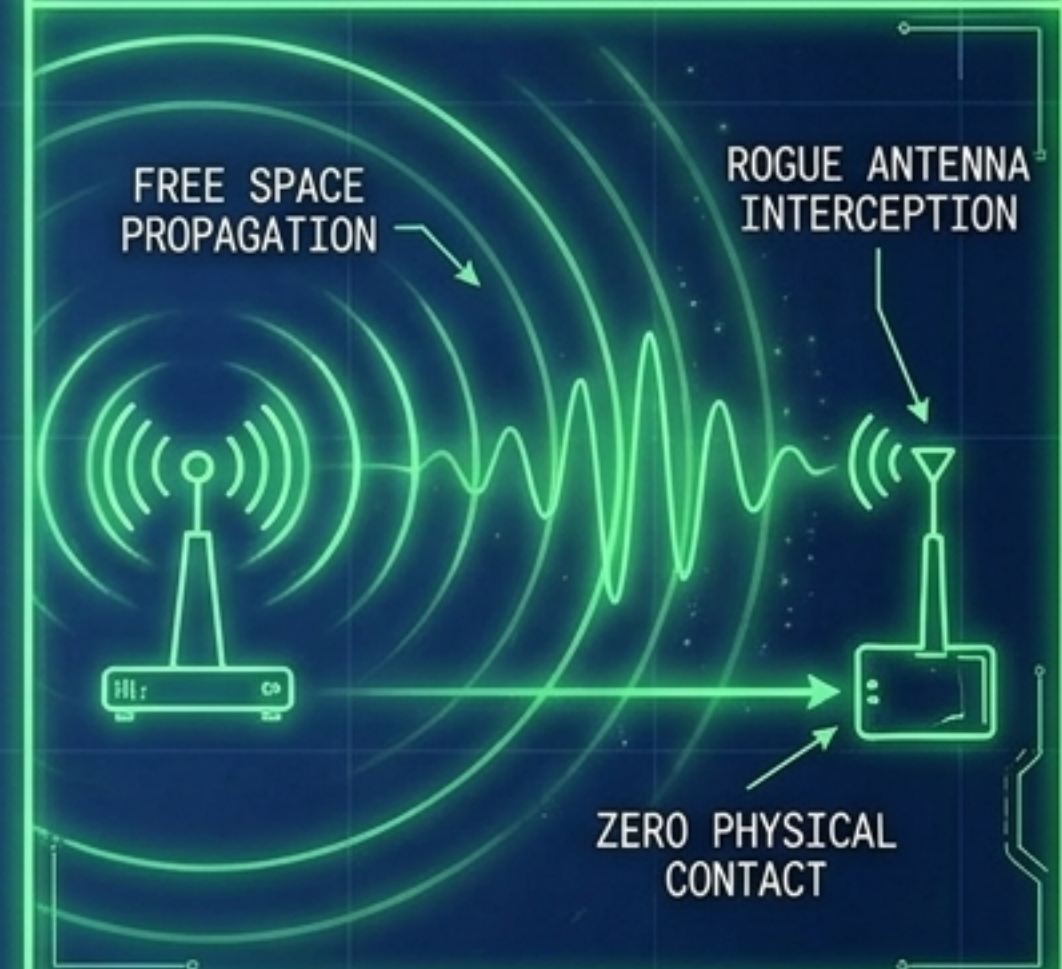
Threat: Highly susceptible to electrical passive tapping if physically exposed.

Fiber (Optical)



Threat: More difficult to tap, but specialized micro-bends can extract light. Not perfectly secure.

Wireless (Radio)



Threat: Inherently public. Interception requires zero physical contact. Attacker only needs to be in range.

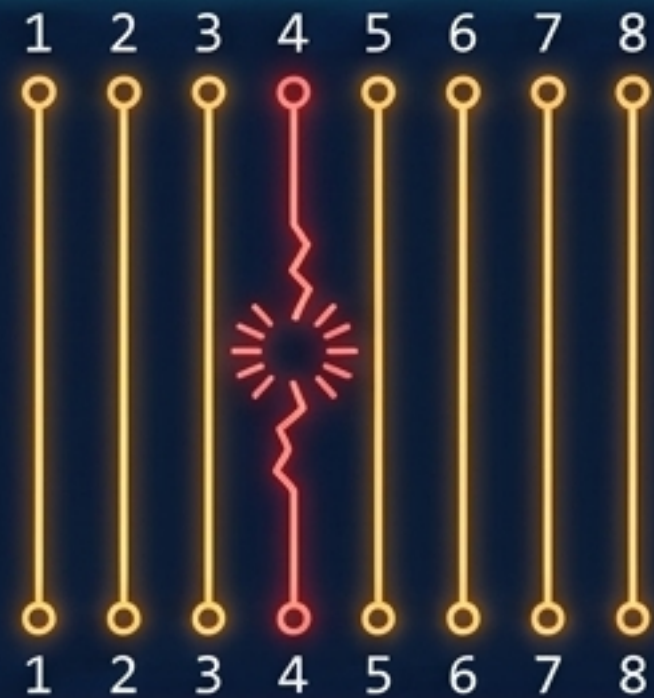
Troubleshooting Layer 1: Link Down vs. Link Slow

Never assume a DNS or IP routing issue before checking Layer 1. A degraded physical link can remain operational while producing disastrous network performance.

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Fault Diagnostic Panel

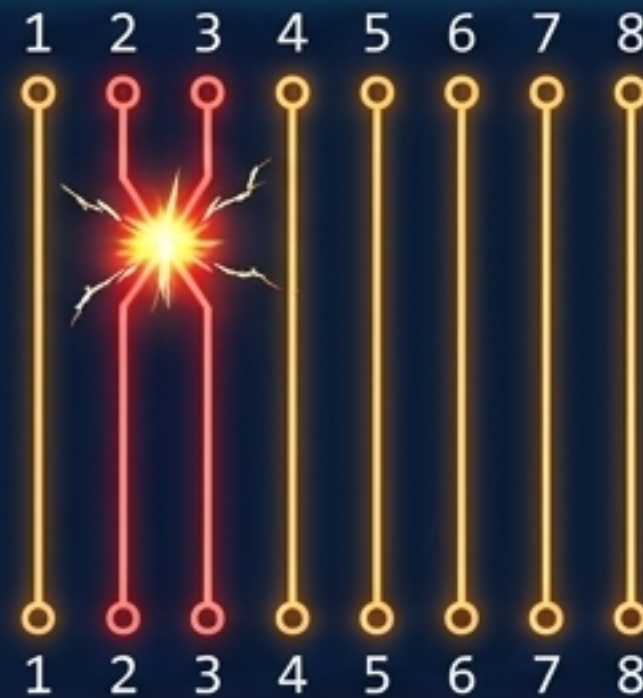
Open Circuit



Result: LINK DOWN

(Cause: Broken conductor, poor termination)

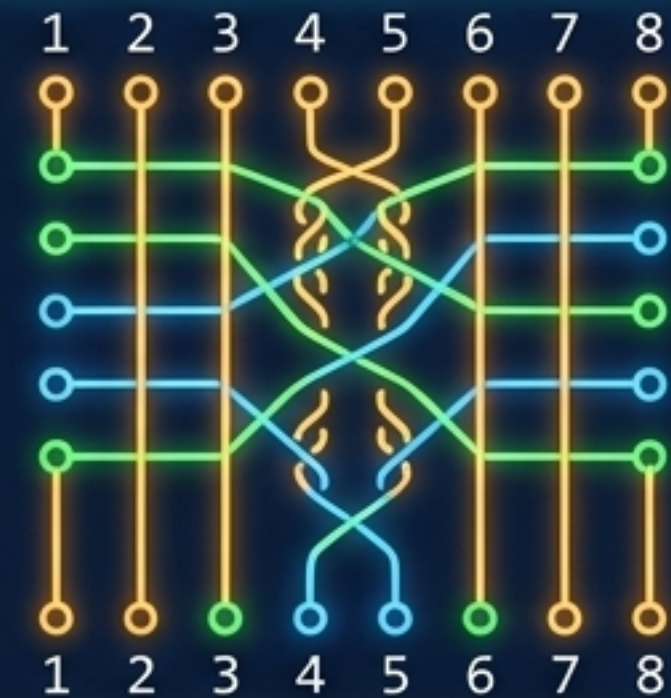
Short Circuit



Result: LINK DOWN

(Cause: Damaged insulation)

Split Pair



Result: LINK SLOW

(Cause: Bad punch-down. Passes basic continuity, fails certification due to severe crosstalk).

The Ultimate Media Selection Matrix

There is no universally best medium. The correct choice is a pure engineering decision balancing bandwidth, distance, environment, and security.

	Twisted Pair	Coaxial	Fiber Optic	Wireless
Signal	Electrical	Electrical/RF	Optical	Radio
EMI Resistance	Limited	Good (Shielded)	Excellent (Immune)	Subject to RF Interference
Shared Medium	Point-to-Point	Yes (DOCSIS)	Point-to-Point	Yes (Airspace)
Interception Risk	Passive Tap	Passive Tap	Optical Splitter	In-Range Capture
Ideal Use Case	Office LAN/Desktops	Broadband Access	WAN/Data Center/ISP	Mobility/Flexibility

The Foundation of the Network Stack

The networking stack rests entirely on physical reality.

Packets cannot travel without signals.
Signals cannot travel without a medium.

The medium defines the absolute limits of the network.

